

Assignment On

‘ADVANCED DATA STRUCTURES AND ALGORITHMS’

(Assignment-1)

Submitted by:

Madhu boini

2503B09904(MCA).

Submitted to:

Dr. Rahul Mishra.

Question:

1. **Implement a Singly Linked List to Manage Patient Details (Name, Age, ID). Perform Insertion and Deletion.**

Code :

class Patient:

def __init__(self, name, age, patient_id):

self.name = name

self.age = age

self.id = patient_id

self.next = None

class PatientList:

def __init__(self):

self.head = None

Insert patient at beginning

def insert_patient(self, name, age, patient_id):

new_patient = Patient(name, age, patient_id)

new_patient.next = self.head

self.head = new_patient

```
print("Patient inserted:", name)

# Delete patient by ID
def delete_patient(self, patient_id):
    temp = self.head
    prev = None

    # If head node itself holds the ID
    if temp is not None and temp.id == patient_id:
        self.head = temp.next
        print("Patient deleted with ID:", patient_id)
        return

    # Search for the ID
    while temp is not None and temp.id != patient_id:
        prev = temp
        temp = temp.next

    # If patient not found
    if temp is None:
        print("Patient not found with ID:", patient_id)
        return

    # Unlink the node
    prev.next = temp.next
    print("Patient deleted with ID:", patient_id)

# Display all patients
def display(self):
    temp = self.head
    if temp is None:
```

```

        print("No patients in the list.")

    return

    print("\n--- Patient List ---")

    while temp is not None:

        print(f"Name: {temp.name}, Age: {temp.age}, ID: {temp.id}")

        temp = temp.next

# -----
# Example Usage
# -----

plist = PatientList()

plist.insert_patient("Ravi", 30, 101)
plist.insert_patient("Sita", 25, 102)
plist.insert_patient("Kiran", 40, 103)

plist.display()

plist.delete_patient(102)

plist.display()

```

Output:

```

Patient inserted: Ravi
Patient inserted: Sita
Patient inserted: Kiran

--- Patient List ---
Name: Kiran, Age: 40, ID: 103
Name: Sita, Age: 25, ID: 102
Name: Ravi, Age: 30, ID: 101
Patient deleted with ID: 102

--- Patient List ---
Name: Kiran, Age: 40, ID: 103
Name: Ravi, Age: 30, ID: 101
PS C:\Users\Madhu\OneDrive\Pictures\Desktop\ADSA\Assignment-1>

```

